Name: ____________________________ Date: ________________ Period: _________

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Scalar _______
2. Magnitude _______
3. Unit vector _______
4. Linear combination _______
5. Span _______
6. Dot product _______
7. Column space _______
8. Rank _______

(A) The number of independent columns of a matrix.

(B) A vector whose length is exactly 1.

(C) All vectors you can reach as $A\mathbf{x}$ — the span of a matrix's columns.

(D) A single number that scales (stretches or flips) a vector.

(E) $\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2$; it measures length and angle.

(F) The length of a vector, $\sqrt{v_1^2 + v_2^2}$.

(G) A sum of scaled vectors, $c\mathbf{v} + d\mathbf{w}$.

(H) The set of *all* linear combinations of a set of vectors.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. Two nonzero vectors are **perpendicular** exactly when (A) their sum is $\mathbf{0}$ (B) $\mathbf{v} \cdot \mathbf{w} = 0$ (C) $\mathbf{v} \cdot \mathbf{w} = 1$ (D) they have equal length
2. The vectors $\left[\begin{smallmatrix}1\\2\end{smallmatrix}\right]$ and $\left[\begin{smallmatrix}3\\6\end{smallmatrix}\right]$ span (A) a single point (B) a line (C) the whole plane (D) all of space
3. The product $A\mathbf{x}$ is (A) a combination of the *columns* of A (B) a combination of the rows of A (C) always $\mathbf{0}$ (D) the length of $\mathbf{x}$
4. The matrix $\left[\begin{smallmatrix}1&2\\2&4\end{smallmatrix}\right]$ has rank (A) 0 (B) 1 (C) 2 (D) 4
5. The unit vector $\dfrac{\mathbf{v}}{\|\mathbf{v}\|}$ has length (A) 0 (B) 1 (C) $\|\mathbf{v}\|$ (D) $\|\mathbf{v}\|^2$
6. A vector $\mathbf{b}$ lies in the column space of A exactly when (A) $\mathbf{b} = \mathbf{0}$ (B) $A\mathbf{x} = \mathbf{b}$ has a solution (C) $\mathbf{b}$ is a unit vector (D) $\mathbf{b}$ is perpendicular to every column

Part C — Short Answer & Computation (30 pts)

Show your work.

1. Let $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute **(a)** $\mathbf{v} + \mathbf{w}$, **(b)** $2\mathbf{v} - \mathbf{w}$, **(c)** $\|\mathbf{v}\|$.

2. Write the unit vector that points the same direction as $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

3. Let $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{u} \cdot \mathbf{v}$, then state whether $\mathbf{u}$ and $\mathbf{v}$ are perpendicular and how you know.

4. Find weights c and d so that $c\begin{bmatrix} 2 \\ 1 \end{bmatrix} + d\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$.

5. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Compute $A\mathbf{x}$ *as a combination of the columns of* A.

6. Factor $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$ as $A = CR$. Which columns are independent? Write C, build the recipe R, and give the rank.

7. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. For each target, say whether it is reachable as $A\mathbf{x}$ (is it in the column space?) and justify in one line. **(a)** $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ **(b)** $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. A matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ has columns $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$. **(a)** Is the column space a line or the whole plane? How do you know? **(b)** Is $\mathbf{b} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ reachable? If so, find the weights. **(c)** If the second column were changed to $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$, what would happen to the column space, and why?

2. Two viewers' taste profiles are $\mathbf{u} = \left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right]$ and $\mathbf{v} = \left[\begin{smallmatrix} -1 \\ 2 \end{smallmatrix}\right]$. Compute $\mathbf{u} \cdot \mathbf{v}$ and explain what the result says about how similar the two viewers' tastes are, using what you know about the sign of a dot product.